

Yun Xing

University of British Columbia, Canada
4th year in Applied Science, Engineering Physics

 LinkedIn  Portfolio
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EXPERIENCE

RF Electrical Engineer Co-op

May - December, 2025

Orbital Research Ltd.

Burnaby, BC

- Characterized RF performance of Low Noise Amplifiers (LNAs) and Low-Noise Block Downconverters (LNBs) in L, S, X, Ka, Ku, and Q bands.
- Conducted advanced measurements of S-parameters, noise figure, phase noise, in-band spurs, P1dB, OIP3, image rejection ratio (IRR), and group delay using Keysight PNA-X and PXA signal analyzers.
- Automated test processes by developing Python scripts with SCPI commands to control power supplies, RF signal generators, and measurement instruments.
- Led the commissioning of UHF and L-band ground stations, covering cable specification, link budget and loss calculations, antenna rotation and tracking systems, real-time beacon and image reception and demodulation, as well as overall receiver sensitivity analysis.
- Familiar with US and EU standards on Electromagnetic Compatibility on industry and multimedia devices, and completed immunity, emission, and Electrical Fast Transient compliance testing.

Undergraduate Research Student on NV-Magnetometry in Diamond

May - December, 2024

Advanced Technologies & Micro Systems, Bosch Research

Stuttgart, Germany

- Researched quantum magnetometry based on Nitrogen-Vacancy (NV) centers in diamond, targeting picoTesla-level magnetic field detection.
- Aligned 523nm laser optical setup including beam splinter, filter, and collimator, to excite NV center's spin state with well defined and controlled energy.
- Developed and implemented test setups for magnetometers in thermal vacuum chambers (TVAC) for space-grade applications.
- Modeled magnetic fields and Zeeman splitting of NV spin states using Python and COMSOL.

Undergraduate Research Student on Neutrino Oscillations

January - April, 2023

Particle Physics Group, TRIUMF

Vancouver, BC

- Researched neutrino oscillations and proton decay using the Photosensor Test Facility (PTF).
- Restored full functionality of the PTF system by waterproofing the laser optical box, optimizing operational code, and calibrating key sensors including the polarizer, magnetometer, and accelerometer.
- Performed data analysis of multiple photomultiplier tube (PMT) types using Python and ROOT.
- Authored standard operating procedures (SOPs) for water purification, optical box handling, and gantry setup, initialization, and motion control.

Undergraduate Research Student on Climate Influence of Volcano Eruptions

May - August, 2022

Institute of Space and Atmospheric Studies, University of Saskatchewan

Saskatoon, SK

- Researched the impact of volcanic eruptions on Earth's radiative energy imbalance and surface temperature variability using satellite remote sensing data.
- Utilized SciPy linear regression to estimate global temperature anomalies from satellite data, incorporating measurement uncertainties and error propagation.
- Interpreted results in the context of volcanic aerosol forcing and its role in short-term climate variability.
- Documented detailed code base environment setup instruction to ensure reproducibility of climate analyses.

PROJECTS

RFIC Design: Low Noise Amplifier (CMOS 45nm)

UBC ELEC 404

- Using Cadence Virtuoso, designed and simulated a fully integrated, inductor-degenerated Common-Source Cascode LNA at center frequency of 5.25GHz
- Achieved a forward gain (S_{21}) of 15.7 dB and a Noise Figure (NF) of 2.12 dB, meeting project specification.
- Implemented output tap capacitor and added C_{gs} at input to achieve input (S_{11}) and output (S_{22}) matching of -10.5 dB and -14.5 dB, respectively, across a ± 100 MHz bandwidth.
- Characterized performance with a third-order input intercept point (IIP3) of -7.0 dBm using two-tone QPSS simulations.
- Minimized total power consumption to 3.7 mW at a 1.0 V supply, significantly exceeding project efficiency targets while maintaining all performance metrics.

Radio-Frequency Communications Team Lead

UBC Orbit Satellite Design Team

- Lead the Radio Communications team to develop a half-duplex UHF link for the 1U LEO Earth Observation CubeSat, ALEASAT.
- Manage project licensing process, communicating with national and international regulatory bodies to secure proper spectrum licensing for launch, and remote sensing license for image capture.
- Perform link budget analysis and verification through ground line-of-sight testing, S-parameter measurements using a VNA, and noise figure and receiver sensitivity evaluations.
- Design and simulate RF components, including matching networks, baluns, and filters, using LTSpice, and analyze antenna radiation patterns in Ansys HFSS.
- Design an RF communications PCB in Altium, managing schematic capture, parts selection, RF routing, and layout. Secured sponsorship from local manufacturers for professional PCB fabrication and assembly.

Student Design Team Co-Captain

UBC Orbit Satellite Design Team

- Co-Lead a multidisciplinary team of 80 students from the University of British Columbia and Simon Fraser University in designing, testing, and operating nano-satellites, securing a launch opportunity with the European Space Agency.
- Directed a three-week environmental test campaigns (vibration, thermal balance, and thermal cycling) for an engineering model at ESA's CubeSat Support Facility in Belgium.
- Manage a \$30,000 annual budget, including prepare funding applications, procurement approval, seeking sponsorships, and expense reporting.
- Conduct project risk analysis and improve progress visualization to enhance stakeholder communication and decision-making.
- Direct outreach initiatives, organizing public seminars and social media campaigns to promote space exploration and showcase the team's achievements.

Autonomous Driving Robot

UBC ENPH 353

- Implemented a motor speed control feedback circuit using difference amplifier, including active speed counting and hexadecimal display.
- Soldered H-Bridges for motor control and used STM32 microcontroller and Pulse Width Modulation output to implement differential rear wheel drive.
- Designed and soldered a reflective optical analog sensor circuit, applying PID control for error correction on front wheel steering.
- Fabricated robot components using waterjet cutting, laser cutting, lathe, and milling tools.
- Extended autonomous functionality in Robot Operating System with Gazebo simulation in Python.
- Trained a classification neural network to recognize license plate characters from image sensor feed. Taught the robot line-following using reinforcement learning.